PROJECT 1: LINEAR REGRESSION
MASM22/FMSN30/FMSN30F/FMSN40: LINEAR AND LOGISTIC REGRESSION (WITH DATA GATHERING), 2026

Peer assessment version: **17.00 on Monday 27 April**
Peer assessment comments: **17.00 on Tuesday 28 April**
Final version: **17.00 on Wednesday 29 April**

Introduction

We want to model the yearly emissions of atmospheric particles with a diameter between 2.5 and 10 µm, PM_{10}, per capita in Sweden's 290 municipalities using data from Statistics Sweden (Statistiska Centralbyrån), `www.scb.se`. All data is from 2021 and located in `Project1data.xlsx`. See Canvas for maps of the locations of municipalities and counties.

Variable namn	Description
`Kommun`	Municipality number and name
`County`	County (Län) number and name
`Part`	Part of Sweden: 1 = Götaland; 2 = Svealand; 3 = Norrland
`Coastal`	Coastal = any sea area within its borders: 0 = Inland; 1 = Coastal
`Fuel`	Yearly fuel consumption in agriculture, forestry, fishing, production, building, public, transports, and other sectors (GWh / 1000 inhabitants)
`Vehicles`	Number of passenger cars, busses and trucks / 1000 inhabitants
`Builton`	Area covered in buldings, roads, etc (= not nature), (hectares / 1000 inhabitants)
`Seniors`	65+ year olds (percentage)
`Higheds`	At least 3 years of post-secondary (eftergymnasial) education (percentage)
`Income`	Median yearly income (1000 SEK)
`GRP`	Gross Regional Product per capita (1000 SEK)
`PM10`	The yearly emission of PM_{10}-particles (metric tonnes / 1000 inhabitants)

Our goal is to model how the yearly emission of PM_{10} particles varies as a function of one or several of the other variables, $\mathbf{x}_i = (x_{i1}, \ldots, x_{ip})$. We will use a linear regression model $Y_i = \mathbf{x}_i\beta + \varepsilon_i$ where the random errors ε_i are assumed to be pairwise independent and $N(0,\ \sigma^2)$. In order to fulfill these model assumptions we will have to use suitable transformations of both the emissions and some of the other variables.

Part 1. Particle emissions and fuel consumption

According to The Swedish Environmental Protection Agency (Naturvårdsverket), domestic transports is the largest source of PM_{10} particles, contribution 40 % of Sweden's emissions, with the second largest source being industry with a further 32 % of the emissions. The main particle sources are wear on tyres, road surfaces and brakes, construction and demolition industry, mostly road construction, and the use of fossil fuels in processes requiring high temperatures.

We might thus expect that the yearly fuel consumption could be used to explain some of the differences in particle emissions between the municipalities.

Lec.2 1(a). Motivate, with the help of suitable residual plots, why we should take the logarithm of the emissions:

$$\ln(\texttt{PM10}) = \beta_0 + \beta_1 x + \varepsilon.$$

Also determine whether we should use $x = \texttt{Fuel}$ or $x = \ln(\texttt{Fuel})$.

Lec.2 1(b). Use the model with $x = \ln(\texttt{Fuel})$ (*Model.1(b)*) and present a table with the β-estimates and their 95 % confidence intervals.

Plot $\ln(\texttt{PM10})$ against $\ln(\texttt{Fuel})$ together with this estimated linear relationship, its 95 % confidence interval and a 95 % prediction interval for future observations.

Also transform the relationship back to $\texttt{PM10} = \ldots$, and plot `PM10` against `Fuel` together with the estimated relationship, its 95 % confidence interval and 95 % prediction interval.

Comment on any problems with the model fit and how this is reflected in the behaviour of the residuals.

Lec.2 1(c). Express how, according to *Model.1(b)*, the expected emission of `PM10` particles would change in a municipality if the yearly fuel consumption would decrease by 10 %. Also calculate a 95 % confidence interval for this change rate.

Also calculate, with 95 % confidence interval, how much a municipality would have to reduce the fuel consumption in order to half its PM_{10} emissions.

Lec.4 1(d). Test if there is a significant linear relationship between the log-PM_{10} emissions and the log-fuel consumption, according to *Model 1(b)*. Report the type of test you use, the null hypothesis, the value of the test statistic and its distribution when the null hypothesis is true (including the degrees of freedom), the P-value of the test and the conclusion.

Lec.6 1(e). Determine how much of the variability in log-PM10 between municipalities this model can explain.

Part 2. Adding more explanatory variables

Lec.4 2(a). Now we turn to the other numerical variables. Plot log-PM_{10} against each of them and determine which of these should also be log-transformed. Any variable where a relative change is more natural than an additive change, e.g. any economic variable, as well as non-negative positively skewed variables are likely candidates.

R-code-tip: First pick out the observation identifier `Kommun` and the numerical variables from Fuel through PM10 with `select()`.

Then transform the resulting data frame with `pivot_longer()` (from the package `tidyr` which is included in the package `tidyverse`) so that all values for variables Fuel through GRP are placed in one long variable with default name `value` and the new variable `name` indicates which original variable each value came from. The `Kommun`- and `M10`-values are repeated a sufficient number of times and retain their old names.

You can then plot PM10 against each of the other numerical variables in separate subplots using `facet_wrap()` where the `"free_x"` option lets `ggplot()` pick different limits on the different x-axes:

```
df |> select(Kommun, Fuel:PM10) |>
  pivot_longer(cols = Fuel:GRP) |>
  ggplot(aes(x = value, y = PM10)) + geom_point() +
    facet_wrap(~ name, scales = "free_x") +
    scale_y_log10()
```

Lec.4 2(b). Plot the (log-transformed) numerical x-variables against each other and determine which of them have strong (at least ± 0.7) pair-wise correlations with each other.

R-code tip: Use `select() |> pivot_longer()` in order to place the x-variables in one long variable as before, and use `mutate()` to get the log-values instead.

Then transform the data back to wide format with `pivot_wider()`. The option `id_cols` is used to indicate which values come from the same municipality, `names_from` denotes where to get the variable names from and `names_prefix` adds a prefix to the original variable names, indicating that we now have log-values, while `values_from` indicates where the log-values come from.

Remove the factor variable `Kommun` (we do not want to plot it) and plot all pairs of log-values with `ggpairs()` from the package `GGally` (you may have to install it first). You also have to add, and run, `library(GGally)` at the start of your code.

```
#library(GGally)
df |> select(Kommun, Fuel:GRP) |>
  pivot_longer(cols = Fuel:GRP) |>
  mutate(value = log(value)) |>
  pivot_wider(id_cols = Kommun, names_from = name,
              names_prefix = "log", values_from = value) |>
  select(-Kommun) |>
  ggpairs()
```

Lec.4 2(c). Ignore any potential problems with multicollinearity and fit a model for log-PM_{10} as a linear function of all seven log-transformed numerical variables and present a table of the β-estimates and their standard errors.

Calculate the VIF-values for the x-variables and add them to the table. Determine whether it is reasonable to use all seven x-variables in the model or if some of them have issues.

Motivate which variable we should exclude first, refit the model without it (*Model 2(c)*) and add the new β-estimates for the remaining x-variables to the table, together with their new standard errors and VIF-values.

Compare the sizes of the standard errors for the β-estimate for the variable that had the second largest VIF-value, before we removed the one with the largest VIF-value. Explain the difference in size between the two standard errors and relate it to the relevant VIF-value.

Lec.4 2(d). Test if there is still a significant, at level $\alpha = 5\,\%$, linear relationship between the log-PM_{10} emissions and the log-fuel consumption, when we have added another five x-variables to the model, according to *Model 2(c)*. Report the type of test you use, the null hypothesis, the value of the test statistic and its distribution when the null hypothesis is true (including the degrees of freedom), the P-value of the test and the conclusion.

Also test if there is a significant, at level $\alpha = 5\,\%$, linear relationship between the log-PM_{10} emissions and at least one of the added five x-variables, according to *Model 2(c)* compared to *Model 1(b)*. Report the type of test you use, the null hypothesis, the value of the test statistic and its distribution when the null hypothesis is true (including the degrees of freedom), the P-value of the test and the conclusion.

Finally, determine how much of the variability in log-PM10 between municipalities *Model 2(c)* can explain.

Lec.4 2(e). Turn the categorical variable `Part` into a factor variable and present the number of observations in each of the three parts.

Add `Part` to *Model 2(c)*, producing *Model. 2(e)*, and present a table with the resulting β-estimates, their standard errors and suitable versions of the corresponding (G)VIF-values, with Df-values.

What is the reference category here? Is this a suitable reference considering the number of observations? Are there any multicollinearity issues?

Test if there are any significant differences in log-PM10 between any of the three parts of Sweden, when we have taken the six numerical variables into account. Report the type of test you use, the null hypothesis, the value of the test statistics and its distribution when the null hypothesis is true (including the degrees of freedom), the P-value of the test and the conclusion.

Lec.4 2(f). Add all two-way interactions between `Part` and each of the six numerical variables to *Model* **??**, producing *Model 2(f)*.

R-code tip: `(log(Fuel) + [etc] + log(GRP))*Part` gives all pairwise interactions involving `Part`.

Test if any of the added interactions is significantly, at level $\alpha = 5\,\%$, different from zero. Report the type of test you use, the null hypothesis, the value of the test statistics and its distribution when the null hypothesis is true (including the degrees of freedom), the P-value of the test and the conclusion.

This large model with everything contains a number of unnecessary variables. Our next step will be to eleminate some of them.

Part 3. Model validation and selection

Lec.5+ Coming soon